

Face Anonymization: Five-Pipeline Benchmark

Accuracy, downstream utility and edge deployability

19 September 2026

Summary

- **Recommended: T2 (SCRFD-2.5G + SegFace).** Second-best on every quality metric, best downstream retention (93.9%), recall 0.828, and the fastest projected pipeline on the RDK X5 (80–100 FPS).
- **T3 (YOLO11n-Face + MobileSAM)** has the best masks (boundary F1 0.755) but the lowest recall of the cascades (0.739) and would be the slowest cascade on the X5 (6–9 FPS).
- **SANSA (semantic, promptable) is not viable here.** It recovers only 52.6% of faces, is 1 GB, and is effectively undeployable on the X5.
- **Downstream person retention** (YOLOX-s, before vs. after blurring): T1 92.7%, T2 93.9%, T3 91.7%, T4 92.7%, SANSA 88.2%.
- **Hardest category: huge close-up faces.** Boundary F1 is lowest here for every cascade (T1 0.28, T2 0.55, T3 0.61, T4 0.27), since a large face’s long outline is judged against a fixed 3 px tolerance.

1 Setup

Pipelines. Four detector + mask cascades: T1 RetinaFace-R50 (detector) + MediaPipe (mask), T2 SCRFD-2.5G (detector) + SegFace (mask), T3 YOLO11n-Face (detector) + MobileSAM (mask), T4 YOLOv5n-Face (detector) + MediaPipe (mask). SANSA is a fifth candidate, a few-shot generalist segmenter (paper-accurate generalist adapter, 1-shot mask prompt) that detects and segments jointly from one prompt, with *no* separate detector — included to test whether one promptable model could replace the four task-specific cascades.

Data. 98 images, 157 annotated faces (COCO polygons), in 7 scenario categories (face count in parentheses): general (8), tiny face (22), huge close-up (14), pose variation (35), occlusion (18), multi-face mixed scale (41), challenging lighting (19).

Protocol. A prediction is a true positive when box IoU ≥ 0.50 and confidence ≥ 0.40 (greedy matching). Mask metrics are computed only over TP-matched faces, so they must be read together with recall. Boundary IoU uses a band of 2% of the face-crop diagonal; boundary F1 uses a fixed 3 px tolerance. Hardware: shared RTX A6000, fp32.

Anonymization. T1, T2 and T4 apply a Gaussian blur inside the mask, with kernel size scaled to the face (12% of the box) and a 3 px feathered edge. T3 uses radial pixelation instead (blocks of 24 px at the centre down to 4 px at the edge). SANSA uses a fixed 51×51 blur with a hard edge. The operators therefore differ, which slightly confounds the retention comparison in §3.

2 Segmentation quality

Table 1: Main results. Mask metrics over TP-matched faces. Best per column in bold.

Pipeline	Recall	TP/GT	Mask IoU	Dice	Bnd. IoU	Bnd. F1	Retention	Size (MB)
T1 RetinaFace-R50 + MediaPipe	0.898	141/157	0.689	0.810	0.122	0.469	92.7%	113.3
T2 SCRFD-2.5G + SegFace	0.828	130/157	0.769	0.862	0.240	0.695	93.9%	88.2
T3 YOLO11n-Face + MobileSAM	0.739	116/157	0.796	0.867	0.328	0.755	91.7%	46.2
T4 YOLOv5n-Face + MediaPipe	0.854	134/157	0.679	0.802	0.121	0.438	92.7%	18.1
SANSA	0.526	82/156*	0.647	0.781	0.162	0.408	88.2%	1003.4

*One image is held out as SANSA’s 1-shot support exemplar.

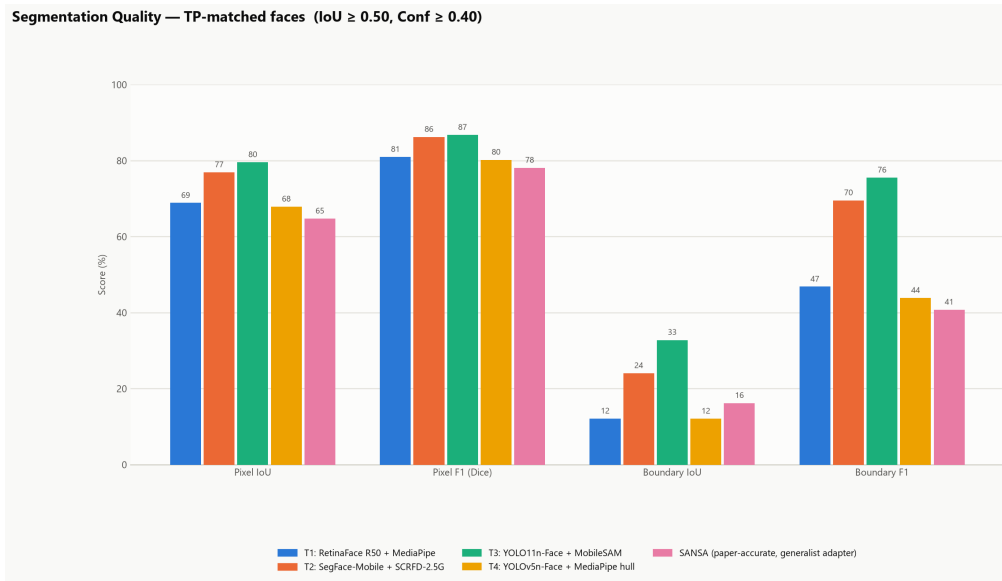


Figure 1: Segmentation quality on TP-matched faces.

Pixel IoU is similar across the cascades (0.68–0.80); **boundary quality is what separates them.** T3 and T2 produce true segmentation masks and lead on boundary F1 (0.755, 0.695). T1 and T4 approximate the face with a MediaPipe landmark hull and fall to 0.44–0.47. SANSA is last or near-last on every metric and finds only half the faces.



Figure 2: Boundary F1 by scenario category.

Per category: huge close-ups are the hardest case for boundaries in every cascade (T1/T4 drop to 0.28/0.27), because a long face outline is judged against a fixed 3 px tolerance. SANSA’s recall collapses on pose variation (0.229), since a single frontal exemplar does not generalize to other poses. On tiny faces only T1 keeps its recall (0.909); T2 and T4 fall to 0.682 and T3 to 0.591.

3 Downstream utility (YOLOX person retention)

Blurring faces should not stop other systems from detecting people. Each pipeline’s own detector + mask model (§1, Table 1) produces the anonymized image; we then run a separate, independent **downstream detection model**, **YOLOX-s** (official COCO weights, 640×640 , conf 0.01, NMS 0.65), on each image before and after that anonymization, keeping only the COCO “**person**” class, since YOLOX has no face class and plays no part in producing the blur. Person boxes are matched before-to-after at $\text{IoU} \geq 0.50$, and retention is the fraction of originally detected people still found, pooled over all 412 people in the dataset.

Table 2: Person retention after anonymization (YOLOX-s).

Pipeline	Persons before	Retained	Retention	Images with a lost person
T1	412	382	92.7%	20 / 98
T2	412	387	93.9%	17 / 98
T3	412	378	91.7%	22 / 98
T4	412	382	92.7%	19 / 98
SANSA	442*	390	88.2%	29 / 97

*SANSA’s retention check ran in a separate environment; compare rates, not counts.

All four cascades lose 6–8% of person detections, so retention barely separates them (T3’s slightly lower figure coincides with its use of pixelation). SANSA is clearly worst at 88.2%, consistent with it also blurring non-face skin.

4 Efficiency and RDK X5 projection

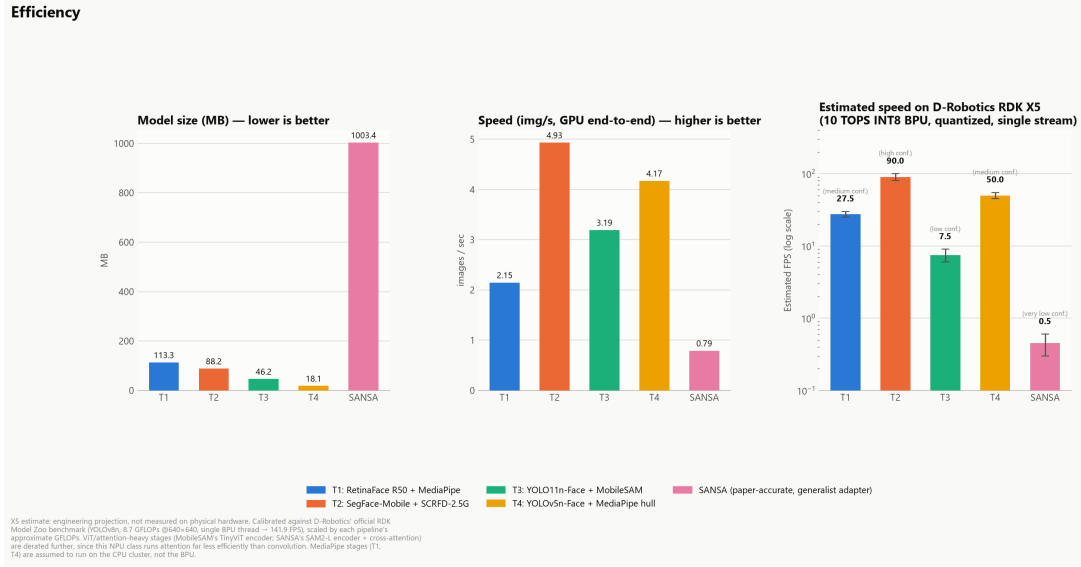


Figure 3: Model size, measured GPU speed, and projected RDK X5 speed (estimate).

How the X5 figures were estimated. The RDK X5 has a 10 TOPS INT8-only NPU and an 8-core Cortex-A55 CPU, so every model would need INT8 quantization. Instead of assuming a fraction of the 10 TOPS peak, we anchored to D-Robotics' published benchmark on this exact board: YOLOv8n (8.7 GFLOPs) runs at 141.9 FPS on one NPU thread. Each pipeline's speed was scaled from that point by its compute cost in GFLOPs, with stages added in sequence.

Two adjustments were made. Transformer (attention) stages were slowed by a further 4–5 $\times$, because this NPU is built for convolutions; this affects MobileSAM (T3) and, heavily, SANSA's SAM2-Large encoder. MediaPipe stages (T1, T4) were assumed to run on the CPU, since MediaPipe is not in the NPU model zoo. These are projections, not measurements, and should be confirmed on hardware.

Table 3: Projected single-stream throughput on the RDK X5.

Pipeline	GPU (img/s)	X5 est. (FPS)	Confidence
T2 SCRFD-2.5G + SegFace	4.93	80–100	high
T4 YOLOv5n-Face + MediaPipe	4.17	45–55	medium
T1 RetinaFace-R50 + MediaPipe	2.15	25–30	medium
T3 YOLO11n-Face + MobileSAM	3.19	6–9	low
SANSA	0.79*	0.3–0.6	very low

*Measured on an RTX 4050 laptop, not the A6000; not directly comparable.

T3 drops from mid-pack on the GPU to slowest cascade on the X5 because of its transformer encoder. SANSA would likely not run on the X5 at all without major rework.

5 Caveats

- SANSA ran on a different machine and software stack, so its latency is not directly comparable; its accuracy numbers are.
- SANSA has no native confidence score; mean mask probability per region was used in its place for the 0.40 threshold. This is unverified and may understate its recall.
- The RDK X5 speeds are projections, not measurements.